

# Task 1: Data Quality and Initial Exploratory Assessment

## Marketing Campaign Performance Analysis

July 28, 2026

## 1 Introduction

Prior to conducting business intelligence analysis, the advertising campaign dataset was evaluated to ensure its quality, reliability, and suitability for analysis. The objective of this assessment is to identify potential data quality issues, validate the performance metrics used throughout the project, examine unusual observations, and establish a set of baseline metrics that will guide subsequent business analysis.

The dataset contains daily advertising campaign performance across multiple platforms, regions, target audiences, product categories, and creative types during the first quarter of 2024.

## 2 Data Quality Assessment

### 2.1 Data Completeness

The dataset was examined for missing values to determine whether incomplete observations could affect subsequent analyses.

- **Revenue**

A sample of these datapoints are shown in *Fig. 1*. It can be seen that twenty observations (approximately 0.6% of the dataset) contained missing revenue values.

2. Revenue

There seems to be some missing values for the `Revenue` column as well. This column is very important as it is used as the basis for the KPI metrics like ROAS, CPA, etc.

```
df[df["Revenue"].isna()].head()
```

|      | Date       | Platform | Campaign                               | Region    | Spend   | CPM   | Impressions | Frequency | Clicks | Purchases | Revenue | Product_Category | Target_Audience | Creative_Type | Video_Completion_Rate | Customer_LTV | Is_Competitor |
|------|------------|----------|----------------------------------------|-----------|---------|-------|-------------|-----------|--------|-----------|---------|------------------|-----------------|---------------|-----------------------|--------------|---------------|
| 194  | 2024-01-06 | Google   | Google_Athletes_Search_Protein_001     | Northeast | 1225.52 | 18.09 | 67745       | 2.27      | 1236   | 35        | NaN     | Protein          | Athletes        | Search        | NaN                   | 940.8        |               |
| 589  | 2024-01-17 | Google   | Google_Athletes_Search_Protein_001     | South     | 978.48  | 15.24 | 64204       | 2.03      | 979    | 32        | NaN     | Protein          | Athletes        | Search        | NaN                   | 940.8        |               |
| 1413 | 2024-02-09 | FB       | FB_FitnessEnth_Carousel_WeightLoss_001 | South     | 1809.27 | 12.47 | 153133      | 3.61      | 823    | 24        | NaN     | WeightLoss       | FitnessEnth     | Carousel      | NaN                   | 448.5        |               |
| 1658 | 2024-02-16 | FB       | FB_Athletes_Video_Protein_001          | Northeast | 974.65  | 12.55 | 77649       | 2.08      | 505    | 16        | NaN     | Protein          | Athletes        | Video         | 0.4413                | 772.8        |               |
| 1661 | 2024-02-16 | FB       | FB_Athletes_Image_PreviewWorkout_002   | South     | 913.94  | 12.55 | 72812       | 4.25      | 243    | 6         | NaN     | PreviewWorkout   | Athletes        | Image         | NaN                   | 538.2        |               |
| 1682 | 2024-02-16 | TT       | TT_Athletes_Video_Protein_001          | Northeast | 920.33  | 10.46 | 87985       | 2.92      | 513    | 20        | NaN     | Protein          | Athletes        | Video         | 0.6728                | 604.8        |               |
| 1738 | 2024-02-18 | FB       | FB_FitnessEnth_Carousel_WeightLoss_001 | Northeast | 2696.87 | 15.09 | 178704      | 2.17      | 881    | 28        | NaN     | WeightLoss       | FitnessEnth     | Carousel      | NaN                   | 448.5        |               |

Figure 1: Sample datapoints with missing revenues.

Because these represented a very small proportion of the data and no systematic pattern of missingness was identified, the corresponding observations were removed as shown in *Fig. 2* prior to analysis.

```

Because the proportion of missing values is very small, removing these observations is unlikely to introduce significant bias or substantially reduce the available information. Consequently, these rows will be removed from the dataset for analyses involving the Revenue variable.

print("BEFORE DROPPING: Number of rows with missing revenue:", len(df[df['Revenue'].isna()]))
df.dropna(subset=['Revenue'], inplace=True)
print("AFTER DROPPING: Number of rows with missing revenue:", len(df[df['Revenue'].isna()]))
0.0
BEFORE DROPPING: Number of rows with missing revenue: 20
AFTER DROPPING: Number of rows with missing revenue: 0

```

Figure 2: Dropped datapoints with missing revenues.

## • Video Completion Rate

A large number of missing values were initially observed in the *Video Completion Rate* variable. Around 62% of the total dataset!

```

df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3240 entries, 0 to 3239
Data columns (total 17 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Date                  3240 non-null  datetime64[ns]
1   Platform              3240 non-null  object
2   Campaign              3240 non-null  object
3   Region                3240 non-null  object
4   Spend                 3240 non-null  float64
5   CPM                   3240 non-null  float64
6   Impressions           3240 non-null  int64
7   Frequency             3240 non-null  float64
8   Clicks                3240 non-null  int64
9   Purchases             3240 non-null  int64
10  Revenue               3220 non-null  float64
11  Product_Category      3240 non-null  object
12  Target_Audience      3240 non-null  object
13  Creative_Type         3240 non-null  object
14  Video_Completion_Rate 1228 non-null  float64
15  Customer_Crv          3240 non-null  float64
16  Is_Competitive_Event  3240 non-null  bool

```

Figure 3: Sample datapoints with missing video\_completion\_rate column

Further investigation revealed that this metric is only applicable to **campaigns using video advertisements**. After restricting the dataframe to video creatives, **only 212 missing observations remained** as shown in *Fig. 4*.

```

df_video_type = df[df['Creative_Type'] == 'Video']
df_video_type.info()
<class 'pandas.core.frame.DataFrame'>
Index: 1440 entries, 0 to 3239
Data columns (total 17 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Date                  1440 non-null  datetime64[ns]
1   Platform              1440 non-null  object
2   Campaign              1440 non-null  object
3   Region                1440 non-null  object
4   Spend                 1440 non-null  float64
5   CPM                   1440 non-null  float64
6   Impressions           1440 non-null  int64
7   Frequency             1440 non-null  float64
8   Clicks                1440 non-null  int64
9   Purchases             1440 non-null  int64
10  Revenue               1424 non-null  float64
11  Product_Category      1440 non-null  object
12  Target_Audience      1440 non-null  object
13  Creative_Type         1440 non-null  object
14  Video_Completion_Rate 1228 non-null  float64
15  Customer_Crv          1440 non-null  float64
16  Is_Competitive_Event  1440 non-null  bool

```

Figure 4: DataFrame filtered to video creative types

Consequently, these missing values were treated as **structurally missing** rather than data quality issues, and analyses involving this variable will use only the relevant subset of campaigns.

## 2.2 Duplicate Records

The dataset was examined for duplicate observations.

```
We checked for duplicated rows and found nothing.

> print("Number of duplicated rows: ", df.duplicated().sum(), "rows")
[238] ✓ 0.0s
... Number of duplicated rows: 0 rows
```

Figure 5: Checking for duplicate records

No duplicate records were identified as shown in *Fig. 5*.

## 2.3 Logical Consistency Checks

Several validation checks were performed to identify logically inconsistent observations.

### 2.3.1 0 Clicks with non-zero purchase

I noticed that there were rows with 0 clicks but has non-zero Purchases, which I thought was weird at first. How are they able to purchase without clicking the ad.

Apparently, this happens in real world scenarios and they are called **view-through conversions**. People see the ad, they don't click it then later that evening, they remembered the brand and typed the website into google and bought the product.

```
df[df['clicks'] == 0]
```

|      | Date       | Platform | Campaign                                  | Region    | Spend   | CPM   | Impressions | Frequency | Clicks | Purchases | Revenue | Product_Category | Target_Audience | Creative_Type | Video_Completion_Rate | Customer_LTV | Is_Competitor |
|------|------------|----------|-------------------------------------------|-----------|---------|-------|-------------|-----------|--------|-----------|---------|------------------|-----------------|---------------|-----------------------|--------------|---------------|
| 51   | 2024-01-02 | Google   | Google_Athletes_Search_Protein_001        | Midwest   | 1045.31 | 15.01 | 0           | 3.09      | 0      | 0         | 2021.74 | Protein          | Athletes        | Search        | NaN                   | 940.8        |               |
| 86   | 2024-01-03 | Google   | Google_Athletes_Search_Protein_001        | Northeast | 1303.24 | 15.03 | 86709       | 2.61      | 0      | 41        | 3093.14 | Protein          | Athletes        | Search        | NaN                   | 940.8        |               |
| 92   | 2024-01-03 | Google   | Google_FitnessEnth_Display_Previewout_001 | West      | 469.88  | 15.03 | 31262       | 2.57      | 0      | 13        | 715.20  | Previewout       | FitnessEnth     | Display       | NaN                   | 545.0        |               |
| 94   | 2024-01-03 | Google   | Google_FitnessEnth_Display_Previewout_001 | Northeast | 921.64  | 15.03 | 61320       | 3.77      | 0      | 12        | 681.21  | Previewout       | FitnessEnth     | Display       | NaN                   | 545.0        |               |
| 122  | 2024-01-04 | Google   | Google_Athletes_Search_Protein_001        | Northeast | 642.88  | 15.04 | 42730       | 3.42      | 0      | 16        | 1388.57 | Protein          | Athletes        | Search        | NaN                   | 940.8        |               |
| ...  | ...        | ...      | ...                                       | ...       | ...     | ...   | ...         | ...       | ...    | ...       | ...     | ...              | ...             | ...           | ...                   | ...          | ...           |
| 3047 | 2024-03-25 | Google   | Google_FitnessEnth_Display_Previewout_001 | Midwest   | 988.78  | 16.26 | 60810       | 3.28      | 0      | 5         | 307.60  | Previewout       | FitnessEnth     | Display       | NaN                   | 545.0        |               |
| 3075 | 2024-03-26 | Google   | Google_Athletes_Search_Protein_001        | Midwest   | 950.82  | 18.72 | 50802       | 2.51      | 0      | 7         | 562.38  | Protein          | Athletes        | Search        | NaN                   | 940.8        |               |
| 3109 | 2024-03-27 | Google   | Google_Athletes_Search_Protein_001        | South     | 1181.40 | 18.73 | 63063       | 2.41      | 0      | 8         | 652.57  | Protein          | Athletes        | Search        | NaN                   | 940.8        |               |

Figure 6: Sample datapoints with 0 clicks but non-zero purchase

But, at this point, I was wondering how they know which ad actually influenced the purchased without click or what if the buyer interacted with multiple ads before purchasing(cross-channel conversions). Apparently, this is related to **attribution**, and there are multiple attribution models like **last\_click attribution**, **first-click attribution**, **linear attribution**, etc.. which is not part of this data cleaning process.

So, for now, I'll just assume that this dataset was collected using a certain attribution model and leave it at that. This justifies the 0 clicks with non-zero purchase.

### 2.3.2 0 Impressions with non-zero Cost-Per-Mile(CPM)

Cost Per Mille (CPM) measures the advertising cost required to generate 1,000 impressions and is commonly used to evaluate the cost efficiency of awareness campaigns.

$$\text{CPM} = \frac{\text{Spend}}{\text{Impressions}} \times 1000 \quad (1)$$

Because impressions appear in the denominator, CPM is undefined when the number of impressions is zero.

During the data quality assessment, 47 observations (approximately 1.45% of the dataset) were identified with **Impressions** = 0 but non-zero CPM values.

Furthermore, all of these observations originated from the Midwest region as shown in *Fig. 7*, suggesting a systematic reporting inconsistency rather than isolated data entry errors.

|     | Date       | Platform | Campaign                               | Region  | Spend  | CPM   | Impressions | Frequency | Clicks | Purchases | Revenue | Product_Category | Target_Audience | Creative_Type | Video_Completion_Rate | Customer_LTV | Is_Competitive |
|-----|------------|----------|----------------------------------------|---------|--------|-------|-------------|-----------|--------|-----------|---------|------------------|-----------------|---------------|-----------------------|--------------|----------------|
| 51  | 2024-01-02 | Google   | Google_Athletes_Search_Protein_001     | Midwest | 845.31 | 15.03 | 0           | 3.09      | 0      | 0         | 2021.74 | Protein          | Athletes        | Search        | NaN                   | 940.80       |                |
| 199 | 2024-01-06 | Google   | Google_WeightLoss_Search_Diet_001      | Midwest | 867.90 | 18.09 | 0           | 4.16      | 0      | 0         | 1542.29 | Diet             | WeightLoss      | Search        | NaN                   | 393.12       |                |
| 251 | 2024-01-07 | TT       | TT_WeightLoss_Video_Diet_001           | Midwest | 619.73 | 12.07 | 0           | 2.11      | 0      | 0         | 3362.89 | Diet             | WeightLoss      | Video         | 0.6101                | 252.72       |                |
| 259 | 2024-01-08 | FB       | FB_Athletes_Image_Protein_002          | Midwest | 745.96 | 12.06 | 0           | 2.83      | 0      | 0         | 527.76  | Protein          | Athletes        | Image         | NaN                   | 538.20       |                |
| 311 | 2024-01-09 | Google   | Google_FitnessEnds_Display_Protein_001 | Midwest | 616.50 | 15.12 | 0           | 2.42      | 0      | 0         | 518.51  | Protein          | FitnessEnds     | Display       | NaN                   | 546.00       |                |

Figure 7: Sample rows that has 0 impressions with non-zero CPM

Since these observations violate the definition of CPM and could affect the calculation of other performance metrics, such as CTR and CVR, they were removed from the dataset prior to analysis.

```

mask = (df["Impressions"] == 0) & (df["CPM"] > 0)

print("Rows before removal:", mask.sum())

df.drop(df[mask].index, inplace=True)

mask = (df["Impressions"] == 0) & (df["CPM"] > 0)
print("Rows after removal:", mask.sum())

```

[356] ✓ 0.0s

... Rows before removal: 47  
Rows after removal: 0

Figure 8: Dropped Rows that has 0 impressions with non-zero CPM

## 2.4 Data Cleaning Summary

The data cleaning process consisted of:

- Removing observations with missing revenue.
- Removing logically inconsistent CPM observations.
- Retaining structurally missing Video Completion Rate values.
- Retaining valid view-through conversions.
- Verifying duplicate observations and variable data types.

Overall, the dataset was determined to be sufficiently complete and reliable for subsequent analysis.

### 3 Performance Metric Validation

To evaluate campaign effectiveness, four standard digital marketing performance metrics were calculated.

| Metric | Formula              |
|--------|----------------------|
| CTR    | Clicks / Impressions |
| CPC    | Spend / Clicks       |
| CVR    | Purchases / Clicks   |
| ROAS   | Revenue / Spend      |

Table 1: Calculated performance metrics.

These metrics provide standardized measures of audience engagement, advertising efficiency, conversion effectiveness, and financial return.

#### Click-Through Rate (CTR)

##### Observation

CTR varies noticeably across campaigns, indicating substantial differences in audience engagement.

##### Why it Matters

Low CTR may indicate ineffective creative design, messaging, or audience targeting and warrants further investigation.

#### Cost Per Click (CPC)

##### Observation

CPC differs considerably between campaigns, suggesting variation in the cost required to generate user engagement.

##### Why it Matters

Campaigns with unusually high CPC may reduce advertising efficiency, particularly if higher acquisition costs do not translate into increased conversions or revenue.

#### Conversion Rate (CVR)

##### Observation

Several campaigns exhibit relatively high CTR but comparatively low CVR.

##### Why it Matters

This suggests that while advertisements successfully attract user attention, they are less effective at converting interest into purchases. Further investigation of post-click experiences is therefore warranted.

#### Return on Advertising Spend (ROAS)

##### Observation

ROAS varies substantially across campaign observations. Several high-spending campaigns generated comparatively low ROAS.

##### Why it Matters

Improving ROAS is the primary business objective. Consequently, identifying the factors associated with low ROAS will be a major focus of the subsequent business analysis.

## 4 Initial Exploratory Assessment

### 4.1 Outlier Assessment

Numerical variables were examined using the Interquartile Range (IQR) method.

Several variables, including advertising spend, impressions, revenue, and customer lifetime value, contained extreme observations. Manual inspection found no evidence that these observations resulted from data entry errors or impossible values. Instead, they represented legitimate campaign outcomes occurring across multiple platforms, regions, and campaign types.

Accordingly, the identified outliers were retained to preserve meaningful variation within the dataset.

### 4.2 Baseline Performance Metrics

Given the business objective of improving advertising efficiency while optimizing budget allocation, the following metrics were selected as the primary indicators of campaign performance.

| Metric  | Business Relevance                                                              |
|---------|---------------------------------------------------------------------------------|
| Spend   | Measures advertising investment and budget allocation.                          |
| Revenue | Measures the financial return generated by advertising activities.              |
| ROAS    | Primary indicator of advertising efficiency and overall campaign profitability. |
| CTR     | Measures audience engagement with advertisements.                               |
| CPC     | Measures the cost required to acquire user engagement.                          |
| CVR     | Measures the effectiveness of converting engaged users into customers.          |

Table 2: Baseline performance metrics used throughout the analysis.

Together, these metrics evaluate campaign performance across the marketing funnel, from advertising investment and user engagement to customer conversion and financial return.

### 4.3 Preliminary Insights

The initial assessment identified several observations that warrant further investigation.

- Campaign performance varies considerably across the dataset.
- High advertising expenditure does not consistently produce higher ROAS.
- Some campaigns successfully attract user engagement but convert relatively few purchases, as indicated by high CTR and comparatively low CVR.
- Other campaigns demonstrate both high engagement and strong conversion performance, suggesting opportunities to identify characteristics associated with successful campaigns.

- These initial findings motivate further analysis by advertising platform, creative type, product category, target audience, geographic region, and competitive events.

## 5 Key Assumptions

The following assumptions were adopted throughout the analysis.

1. Missing Video Completion Rate values are structural because the metric is only applicable to video advertisements.
2. View-through conversions represent legitimate customer behavior and are therefore retained.
3. Campaign performance comparisons should be based on aggregated campaign metrics rather than averages of daily ratios.
4. Extreme observations represent genuine campaign performance unless evidence indicates otherwise.

## 6 Conclusion

The data quality assessment indicates that the advertising campaign dataset is suitable for subsequent business analysis following a limited number of cleaning operations. Missing values and logical inconsistencies were addressed appropriately, while valid observations representing legitimate campaign behavior were retained.

The calculated performance metrics establish a consistent framework for evaluating advertising effectiveness. The preliminary assessment also highlights meaningful variation in campaign performance, particularly regarding engagement, conversion, and advertising return. These findings provide the foundation for the next phase of the project, where business intelligence techniques will be used to investigate campaign performance across multiple business dimensions and develop data-driven recommendations.